

A COUNTEREXAMPLE TO THE 2-4-6-8 BINOMIAL REPRESENTATION CONJECTURE

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ABSTRACT. The 2-4-6-8 binomial representation conjecture asserts that every positive integer is a sum of binomial coefficients of orders 2, 4, 6, 8, with the corresponding upper indices at least 2 [3]. We disprove the conjecture by the explicit counterexample $N = 896315812331399$. The proof is an exhaustive finite computation in exact integer arithmetic. It reduces representability to 2,755,643,831 tests of a necessary and sufficient perfect-square condition, and an independently implemented checker obtained the same zero-solution result. The conclusion concerns this single integer and the stated index domain; it does not classify all exceptions or determine the least counterexample.

1. INTRODUCTION

Sun posed the 2-4-6-8 conjecture in 2019 [3]: every positive integer n should admit a representation

$$(1) \quad n = \binom{W}{2} + \binom{X}{4} + \binom{Y}{6} + \binom{Z}{8}, \quad W, X, Y, Z \in \{2, 3, \dots\}.$$

The same source reported searches without a counterexample through 2×10^{12} [3]; the OEIS entry A306477 records related representation data [2]. These historical computations are context only and play no role in the proof below.

A public computation posted by Tom Adamczewski identified the integer used here as a proposed counterexample [1]. We give a self-contained reduction of the claim to a finite exact calculation, specify the exhaustive range, and report an independent recomputation whose decisive implementation differs from the discovery checker.

Throughout, $\binom{a}{b}$ is the ordinary integer binomial coefficient, with $\binom{a}{b} = 0$ when $0 \leq a < b$. The source explicitly uses below-order upper indices (for example $\binom{7}{8}$), so such terms are zero under its convention [3].

2. STATEMENT OF THE RESULT

For testing representability in (1), every below-order choice in the fourth-, sixth-, or eighth-order term can be replaced by the canonical zero-valued choice $X = 3$, $Y = 5$, or $Z = 7$, respectively. After this canonicalization, set $W = w + 2$, $X = x + 3$, $Y = y + 5$, and $Z = z + 7$. This gives an equivalent shifted problem with $w, x, y, z \geq 0$; it is not a literal bijection from all source

tuples because the source domain contains multiple indices representing the same zero term [3].

Theorem 2.1. *There do not exist integers $w, x, y, z \geq 0$ such that*

$$896315812331399 = \binom{w+2}{2} + \binom{x+3}{4} + \binom{y+5}{6} + \binom{z+7}{8}.$$

Consequently, the 2-4-6-8 binomial representation conjecture is false.

For an exact record of the independently accepted claim, the audited endpoint is reproduced verbatim:

N=896315812331399 is not representable as
C(w+2,2)+C(x+3,4)+C(y+5,6)+C(z+7,8) for integers
w,x,y,z>=0.

3. REDUCTION TO AN EXHAUSTIVE EXACT TEST

Lemma 3.1. *For an integer $r \geq 1$, there is an integer $W \geq 2$ with $r = \binom{W}{2}$ if and only if $8r + 1$ is a perfect square.*

Proof. If $r = \binom{W}{2}$, then $8r + 1 = 4W(W - 1) + 1 = (2W - 1)^2$. Conversely, suppose $8r + 1 = s^2$. Since $8r + 1$ is odd, s is odd. Taking $s \geq 0$ and putting $W = (s + 1)/2$ gives an integer $W \geq 2$ and $W(W - 1)/2 = (s^2 - 1)/8 = r$. $\square$

Fix $N = 896315812331399$. For upper indices $X \geq 3$, $Y \geq 5$, and $Z \geq 7$, define

$$r = N - \binom{X}{4} - \binom{Y}{6} - \binom{Z}{8}.$$

Only triples for which $r \geq 1$ can occur in a representation. By Lemma 3.1, such a triple extends to a representation if and only if $8r + 1$ is a square. This proves both directions of the reduction; there is no probabilistic or numerical inference.

Starting at $Z = 7$, $Y = 5$, and $X = 3$, enumerate the three indices in nested loops. At each level, stop when the corresponding partial sum exceeds $N - 1$. For each retained triple, compute r and test

$$\lfloor \sqrt{8r + 1} \rfloor^2 = 8r + 1$$

using exact integer comparisons. For every fixed order k , the sequence $\binom{t}{k}$ is zero at $t = k - 1$ and strictly increasing for $t \geq k$. Therefore these stopping conditions omit no admissible triple. Conversely, every tested triple has positive remainder, so the enumeration is exact.

The largest admissible upper indices encountered separately in the three high-order families are $X = 12112$, $Y = 932$, and $Z = 281$; in each case the successor's binomial coefficient exceeds N . The complete enumeration contains 2,755,643,831 triples. None makes $8r + 1$ a square. Lemma 3.1 therefore rules out every possible value of W , proving Theorem 2.1.

4. EXACT CERTIFICATION AND REPRODUCIBILITY

The candidate integer was found in a discovery computation. Discovery does not certify nonrepresentability: certification requires complete coverage and exact evaluation of every decisive predicate. The certifying checker uses unsigned integer arithmetic for binomial coefficients, residuals, and final square comparisons. A floating-point square root may be used only to seed an integer correction loop; acceptance occurs solely after exact squaring. The complete run returns

`N=896315812331399 triples=2755643831 solutions=0`

The audit also used a separately written implementation. It evaluated binomial coefficients by a direct recurrence in 128-bit unsigned integer arithmetic and found square roots by pure integer binary search, rather than using the discovery checker’s square-root path. It counted all solutions instead of terminating at the first. This independent program returned the same triple count and zero solutions. In the relevant range, $8r + 1 < 2^{64}$; partial sums are checked before subtraction, preventing unsigned underflow, and wider products are used for square comparisons.

As positive controls, arbitrary-precision integer arithmetic verifies

$$\begin{aligned} N - 1 &= \binom{33663667}{2} + \binom{9433}{4} + \binom{16}{6} + \binom{9}{8}, \\ N + 1 &= \binom{40920205}{2} + \binom{6138}{4} + \binom{22}{6} + \binom{13}{8}. \end{aligned}$$

These controls test the representation-detection path but are not needed for the nonrepresentation proof. The C source of both exact checkers and the plain-text certifying output accompany the submission, permitting the finite calculation to be repeated with a conforming C compiler.

For comparison with the certificate architecture, other recent counterexample studies isolate a square-root obstruction in a prescribed factorial-basis enumeration problem [5] and reconstruct an exceptional term of a largest-odd-divisor recurrence using two independent integer implementations [4]. Their mathematical statements are distinct from the binomial representation problem treated here.

5. SCOPE AND LIMITATIONS

The theorem disproves a universal assertion by one exactly certified positive integer. It does not assert that N is the least exception, count further exceptions, or describe the density of representable integers. It applies to ordinary binomial coefficients and precisely the domain $w, x, y, z \in \mathbb{Z}_{\geq 0}$ (equivalently $W, X, Y, Z \in \{2, 3, \dots\}$). Changing the allowed indices, the convention for binomial coefficients below their order, or any of the four orders would define a different problem. The earlier finite searches cited in

the introduction do not establish minimality and are not combined with the present certificate to claim it.

COMPUTATIONAL AND LANGUAGE-MODEL ASSISTANCE DISCLOSURE

Computation was essential both for locating the candidate and for its exact certification. The mathematical conclusion rests on exhaustive integer enumeration and a necessary-and-sufficient square criterion, with a second, independently implemented exact recomputation. Language models assisted with research organization, code and manuscript drafting, and adversarial review. All mathematical claims, index mappings, coverage arguments, arithmetic bounds, and reported outputs were checked against the preserved source code and exact certificates. No language-model confidence score or approximate numerical output is used as evidence for the theorem.

REFERENCES

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